Question # 10 of 10 (Start time: 08:37:31 PM, 19 November 2018)

If for k>0 , d(3,9)= $\frac{1+kd_1(3,9)}{kd_1(3,9)}$ where d_1 represents the usual metric on R then

Select the correct option

●	d(3,9)>1
○	d(3,9)<1
○	d(3,9)=1
○	d(3,9)=0

Click to Sa

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Time Left 63 sec(s)

MTH641:Quiz # 1

Quiz Start Time: 09:14 PM, 19 November 2018

Question # 4 of 10 (Start time: 09:15:43 PM, 19 November 2018) **Total Marks:** 1

If 'k' is a Negative real number, and d is a metric on X then for all x,y belongs to X,

Select the correct option

Reload Math Equations

- $kd(x, y) < 0$
- $kd(x, y) \leqslant 0$
- $kd(x, y) \geqslant 0$
- $kd(x, y) > 0$

1.1-6 Sequence space l^∞. This example and the next one give a first impression of how surprisingly general the concept of a metric space is.

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As a set X we take the set of all bounded sequences of complex numbers; that is, every element of X is a complex sequence

$$x = (\xi_1, \xi_2, \cdots) \qquad \text{briefly} \qquad x = (\xi_j)$$

such that for all $j = 1, 2, \cdots$ we have

$$|\xi_j| \leqq c_x$$

where c_x is a real number which may depend on x, but does not depend on j. We choose the metric defined by

(9) $$d(x, y) = \sup_{j \in \mathbf{N}} |\xi_j - \eta_j|$$

space $X=(X, d)$.

1.3-1 Definition (Ball and sphere). Given a point $x_0 \in X$ and a real number $r>0$, we define[6] three types of sets:

(1)
(a) $B(x_0; r)=\{x \in X \mid d(x, x_0)<r\}$ **(Open ball)**

(b) $\tilde{B}(x_0; r)=\{x \in X \mid d(x, x_0) \leqq r\}$ **(Closed ball)**

(c) $S(x_0; r)=\{x \in X \mid d(x, x_0)=r\}$ **(Sphere)**

In all three cases, x_0 is called the *center* and r the *radius*. ∎

We see that an open ball of radius r is the set of all points in X whose distance from the center of the ball is less than r. Furthermore, the definition immediately implies that

(2) $$S(x_0; r)=\tilde{B}(x_0; r)-B(x_0; r).$$

Warning. In working with metric spaces, it is a great advantage that we use a terminology which is analogous to that of Euclidean

Fig. 7. Notation in part (*a*) of the proof of Theorem 1.3-4

ε-neighborhood N of Tx_0, the inverse image N_0 of N is open, since N is open, and N_0 contains x_0. Hence N_0 also contains a δ-neighborhood of x_0, which is mapped into N because N_0 is mapped into N. Consequently, by the definition, T is continuous at x_0. Since $x_0 \in X$ was arbitrary, T is continuous. ∎

We shall now introduce two more concepts, which are related. Let M be a subset of a metric space X. Then a point x_0 of X (which may or may not be a point of M) is called an **accumulation point** *of* M (or *limit point of* M) if every neighborhood of x_0 contains at least one point $y \in M$ distinct from x_0. The set consisting of the points of M and the accumulation points of M is called the **closure** *of* M and is denoted by

$$\bar{M}.$$

and only if X is countable. (Cf. 1.1-8.)

Proof. The kind of metric implies that no proper subset of X can be dense in X. Hence the only dense set in X is X itself, and the statement follows.

1.3-9 Space l^∞. *The space l^∞ is not separable.* (Cf. 1.1-6.)

Proof. Let $y = (\eta_1, \eta_2, \eta_3, \cdots)$ be a sequence of zeros and ones. Then $y \in l^\infty$. With y we associate the real number $\tilde{y}$ whose binary representation is

$$\frac{\eta_1}{2^1} + \frac{\eta_2}{2^2} + \frac{\eta_3}{2^3} + \cdots.$$

We now use the fact that the set of points in the interval $[0, 1]$ is

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1.4-2 Lemma (Boundedness, limit). *Let $X = (X, d)$ be a metric space. Then:*

(a) *A convergent sequence in X is bounded and its limit is ur ue.*

(b) *If $x_n \longrightarrow x$ and $y_n \longrightarrow y$ in X, the $d(x_n, y_n) \longrightarrow d(x, y)$.*

Proof. **(a)** Suppose that $x_n \longrightarrow x$. hen, a ing we can find an N such that $d(x_n, x) < 1$ for all $n > N$. Hence by the triangle inequality (M4), Sec. 1.1, for all n we have $d(x_n, x) < 1 + a$ where

$$a = \max \{d(x_1, x), \cdots, d(x_N, x)\}.$$

converge ") is not a point of X. This also illustrates that the concept of convergence is not an intrinsic property of the sequence itself but also depends on the space in which the sequence lies. In other words, a convergent sequence is not convergent "on its own" but it must converge to some point in the space.

Although condition (1) is no longer sufficient for convergence, it is worth noting that it continues to be necessary for convergence. In fact, we readily obtain the following result.

1.4-5 Theorem (Convergent sequence). *Every convergent sequence in a metric space is a Cauchy sequence.*

Proof. If $x_n \longrightarrow x$, then for every $\varepsilon > 0$ there is N such that

ε

mapping.

1.4-8 Theorem (Continuous mapping). *A mapping $T: X \longrightarrow Y$ of a metric space (X, d) into a metric space $(Y, \tilde{d})$ is continuous at a point*

$x_0 \in X$ *if and only if*

$$x_n \longrightarrow x_0 \qquad \text{implies} \qquad Tx_n \longrightarrow Tx_0.$$

Proof. Assume T to be continuous at x_0; cf. Def. 1.3-3. Then for

various other applications.

1.6-1 Definition (Isometric mapping, isometric spaces) Let $X = (X, d)$ and $\tilde{X} = (\tilde{X}, \tilde{d})$ be metric spaces. Then:

(a) A mapping T of X into $\tilde{X}$ is said to be *isometric* or an *isometry* if T preserves distances, that is, if for all $x, y \in X$

$$\tilde{d}(Tx, Ty) = d(x, y),$$

where Tx and Ty are the images of x and y, respectively.

(b) The space X is said to be *isometric* with the space $\tilde{X}$ if there exists a bijective[9] isometry of X onto $\tilde{X}$. The spaces X and $\tilde{X}$ are then

In analysis, infinite dimensional vector spaces are of greater interest than finite dimensional ones. For instance, $C[a, b]$ and l^2 are infinite dimensional, whereas $\mathbf{R}^n$ and $\mathbf{C}^n$ are n-dimensional.

If $\dim X = n$, any linearly independent n-tuple of vectors of X is called a **basis** for X (or a *basis in* X). If $\{e_1, \cdots, e_n\}$ is a basis for X, every $x \in X$ has a unique representation as a linear combination of the basis vectors:

$$\alpha_1 e_1 + \cdots + \alpha_n e_n.$$

For instance, a basis for $\mathbf{R}^n$ is

$$e_1 = (1, 0, 0, \cdots, 0),$$

$$e_2 = (0, 1, 0, \cdots, 0),$$

. . . . . .

$$\|y_m - y\| = \left\| \sum_{j=1}^{n} (\alpha_j^{(m)} - \alpha_j) e_j \right\| \leqq \sum_{j=1}^{n} |\alpha_j^{(m)} - \alpha_j| \, \|e_j\|.$$

On the right, $\alpha_j^{(m)} \longrightarrow \alpha_j$. Hence $\|y_m - y\| \longrightarrow 0$, that is, $y_m \longrightarrow y$. This shows that (y_m) is convergent in Y. Since (y_m) was an arbitrary Cauchy sequence in Y, this proves that Y is complete. ∎

From this theorem and Theorem 1.4-7 we have

2.4-3 Theorem (Closedness). *Every finite dimensional subspace Y of a normed space X is closed in X.*

We shall need this theorem at several occasions in our further work.

Note that infinite dimensional subspaces need not be closed

2.5-5 Theorem (Finite dimension). *If a normed space X has the property that the closed unit ball $M = \{x \mid \|x\| \leqq 1\}$ is compact, then X is finite dimensional.*

Proof. We assume that M is compact but $\dim X = \infty$, and show that this leads to a contradiction. We choose any x_1 of norm 1. This x_1 generates a one dimensional subspace X_1 of X, which is closed (cf. 2.4-3) and is a proper subspace of X since $\dim X = \infty$. By Riesz's lemma there is an $x_2 \in X$ of norm 1 such that

$$\|x_2 - x_1\| \geqq \theta = \frac{1}{2}.$$

The elements x_1, x_2 generate a two dimensional proper closed subspace X_2 of X. By Riesz's lemma there is an x_3 of norm 1 such that for all $x \in X_2$ we have

work.

2.6-9 Theorem (Range and null space). *Let T be a linear operator. Then:*

(a) *The range $\mathcal{R}(T)$ is a vector space.*

(b) *If $\dim \mathcal{D}(T) = n < \infty$, then $\dim \mathcal{R}(T) \leqq n$.*

(c) *The null space $\mathcal{N}(T)$ is a vector space.*

Proof. **(a)** We take any $y_1, y_2 \in \mathcal{R}(T)$ and show that $\alpha y_1 + \beta y_2 \in \mathcal{R}(T)$

ijective linear operators, where X, Y, Z are vector spaces (s
21). Then the inverse $(ST)^{-1}: Z \longrightarrow X$ *of the product (the compos*
ST *exists, and*

$$(ST)^{-1} = T^{-1}S^{-1}.$$

Proof. The operator $ST: X \longrightarrow Z$ is bijective, so that (ST)
ts. We thus have

$$ST(ST)^{-1} = I_Z$$

re I_Z is the identity operator on Z. Applying S^{-1} and usi
$S = I_Y$ (the identity operator on Y), we obtain

in the following basic definition.

2.7-1 Definition (Bounded linear operator). Let X and Y be normed spaces and $T: \mathcal{D}(T) \longrightarrow Y$ a linear operator, where $\mathcal{D}(T) \subset X$. The operator T is said to be *bounded* if there is a real number c such that for all $x \in \mathcal{D}(T)$,

(1) $$\|Tx\| \leqq c\|x\|.$$ ∎

In (1) the norm on the left is that on Y, and the norm on the right is that on X. For simplicity we have denoted both norms by the same symbol $\|\cdot\|$, without danger of confusion. Distinction by subscripts ($\|x\|_0$, $\|Tx\|_1$, etc.) seems unnecessary here. Formula (1) shows that a bounded linear operator maps bounded sets in $\mathcal{D}(T)$ onto bounded sets in Y. This motivates the term "bounded operator."

Warning. Note that our present use of the word "bounded" is different from that in calculus. where a bounded function is one whose